

Cephalometric Norms for Orthognathic Surgery in Black American Adults

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Normal values for the Cephalometrics for Orthognathic Surgery (COGS) analysis for American black adults were developed in this study. The cephalometric radiographs of 33 black American adults with ideal dentitions were analyzed, and a statistical description of their hard and soft tissue cephalometric measurements is presented. In addition, mean values for black American adults were compared with those of white American adults. In black subjects, there was greater maxillary skeletal prognathism, skeletal lower face height, skeletal facial convexity, lower incisor proclination, anterior dental heights, upper and lower lip lengths, and soft tissue thickness of the lips and chin than in white subjects. There was less nasal depth and projection, less bony chin depth, and a smaller nasolabial angle in black subjects. The results of this study may be useful in providing racially specific cephalometric values for diagnosis and treatment planning for orthognathic surgery in American black adults.

The Cephalometrics for Orthognathic Surgery (COGS) analysis of Burstone and colleagues^{1,2} is especially adapted to the diagnosis and treatment planning of orthognathic surgery cases because it is based largely upon rectilinear measurements that can be used at surgery. Normal values for cephalometric analysis of dental and facial form have been extensively developed for North American and Northwestern European populations.³⁻⁸ Normal values for the COGS analysis of black American adults would be useful in providing racially specific values for diagnosis and treatment planning for or-

thognathic surgery. However, such data are not currently available.

Altamus^{9,10} studied a group of 80 American black children and developed norms for the Downs analysis. He found a greater absolute size of black children's heads and greater dental protrusion, with the chin in the same position relative to the cranial base as in white children. In addition, he found that the lower facial height was relatively longer than the upper facial height compared with white children. Altamus further studied the integumental profile of black children and compared his results with those of Burstone in white subjects.¹¹ He found, in general, a greater soft tissue thickness in black children than in Burstone's sample of 37 adults with average faces, except at subnasale, where the soft tissue coverage over the anterior nasal spine was thinner.

Cotton et al¹² performed the Downs¹³ analysis on a small sample of 20 black adults from the San Francisco area. Drummond¹⁴ studied 40 American blacks ranging in age from 8 to 23 years, establishing normal values for the Tweed,¹⁵ Steiner,¹⁶ and Riedel¹⁷ analyses. In comparison with white adult norms, he found a greater mandibular plane angle, greater maxillary skeletal prognathism, and greater upper and lower incisor proclination and interincisal angle.

Kowalski et al¹⁸ studied a population of 244 black

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men, ranging in age from 20 to 60 years, who were not selected for ideal dental occlusion. They presented mean values for the Steiner analysis.¹⁶ Fonseca and Klein¹⁹ studied a sample of 40 black women, aged 20 to 30 years, with Angle class I dental occlusion, good health, and no obvious facial deformities. They presented norms for a variety of measurements from Steiner,¹⁶ Riedel,¹⁷ Downs,¹³ and Burstone⁷ analyses but did not use a rectilinear coordinate system. In comparison to a control sample of 20 white women, they found that the maxilla and mandible had greater skeletal prognathism, greater upper and lower incisor proclination, shorter middle face height, and longer lower face height. Lip protrusion was greater, yet absolute lip thickness was not significantly greater in black than in white women.

Connor and Moshiri²⁰ compared a sample of 50 black adults with a sample of 50 white adults, both groups having Angle class I occlusion. They presented norms for a variety of measurements from several analyses, as did Fonseca and Klein.¹⁹ They found greater maxillary and mandibular skeletal prognathism, anterior dental height, lower incisor proclination, upper lip length, and throat length in blacks than in whites. Nasolabial angle and lip-chin-throat angle were less in blacks than in whites. Male and female norms were presented.

The purpose of this study was to develop normal values for COGS analysis in black American adults.

Materials and Methods

Inactive dental records from the University of Pittsburgh School of Dental Medicine were reviewed as part of a cooperative study with the University of Connecticut School of Dental Medicine. Among them were found 33 black American adult patients who met the criteria of this study. There were 15 men, ranging in age from 18 to 48 years, and 18 women, ranging in age from 17 to 48 years. Women were required to be at least 17 years of age, and men were required to be at least 18 years of age, in order to ensure that facial skeletal maturity had been reached. The average age of the group was 25.8 years. The dental occlusion, as indicated on the occlusal analysis page of the dental chart, was required to be an ideal Angle class I incisor and molar occlusion, without deep bite, open bite, or significant dental rotations or crowding. All teeth anterior to the second molars had to be present. The physical examination page could not have any mention of facial asymmetry or disfigurement. The medical and dental history could have no mention of facial trauma, or orthodontics, or maxillofacial surgery.

The cephalometric radiographs taken from the charts selected according to these criteria were used for analysis. They had been taken on the cephalostat at the University of Pittsburgh School of Dental Medicine, Dental Radiology Clinic, between 1970 and 1977. On some of the radiographs, certain soft tissue structures, such as the tip of the nose or the chin, did not register on the film. Measurements relating to missing structures were deleted from the analysis for that patient. All of the cephalometric radiographs used in this study had an adequate representation of all of the hard tissue structures used in the cephalometric analysis.

Tracings of the cephalometric radiographs were made by hand on 0.003-in thick matte finish acetate sheets (Unitek, Monrovia, NY). The tracings were reviewed on two separate occasions each by two of the investigators (T.R.F. and R.I.A.) for accurate landmark identification. Strict definitions of the cephalometric landmarks used in this study were made in consultation with Dr Burstone and with reference to publications on the COGS analysis.^{1,2} Each tracing was digitized twice on a Tektronix 4953 digitizing tablet (Tektronix, Beaverton, OR) using a computer program developed by Burstone.

Figure 1 illustrates the 51 points digitized on each cephalometric tracing. Figures 2 and 3 demonstrate some of the hard tissue measurements taken from

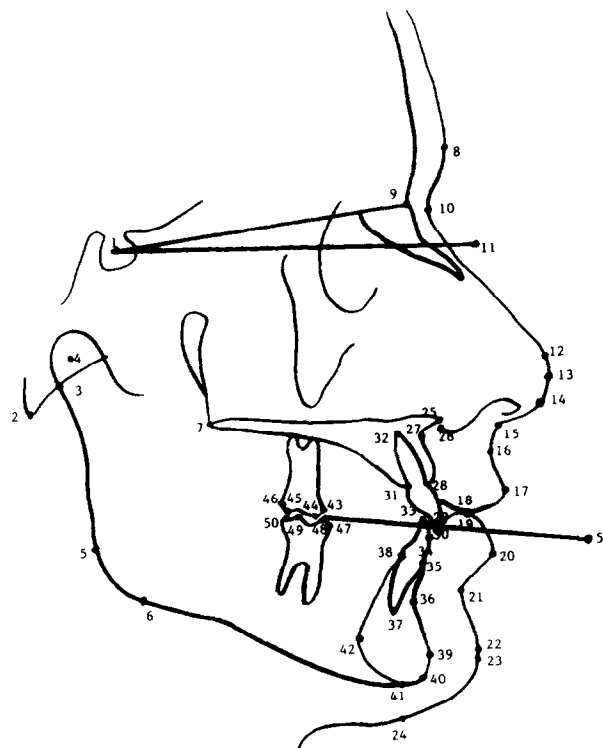
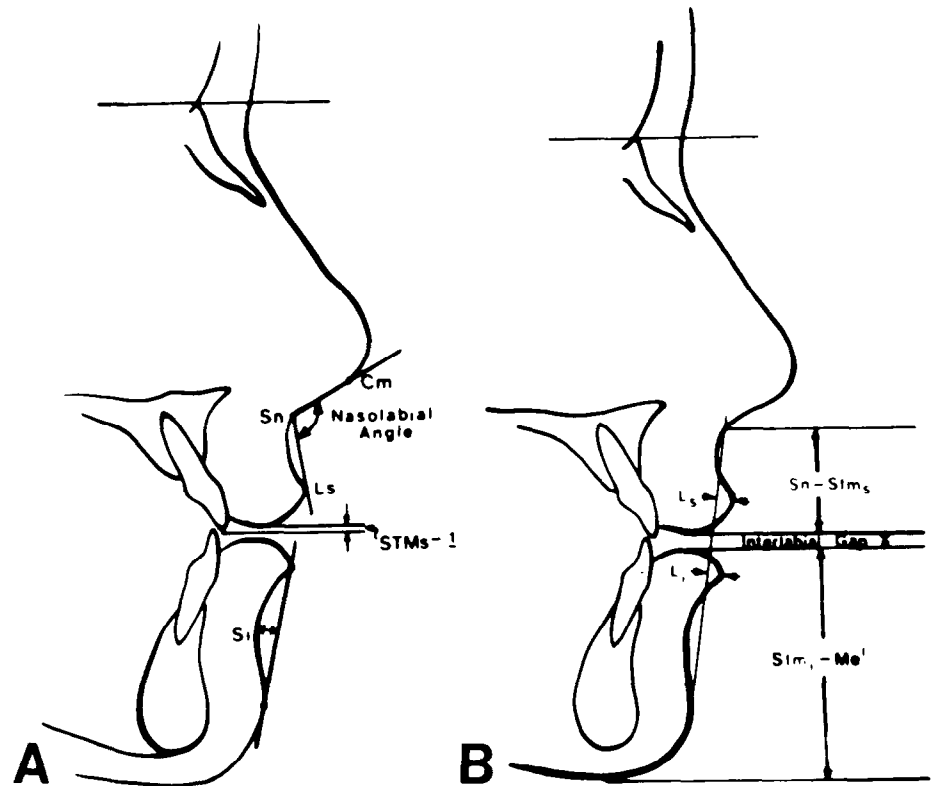


FIGURE 1. The 51 points digitized in the computerized analysis used in this study.

FIGURE 5. Measurements of labial form. A, nasolabial angle, upper incisor display, and mentolabial sulcus. B, upper and lower lip protrusion (arrows), interlabial gap, and lower-third facial height proportionality. Reprinted with permission.²



Therefore, a high confidence interval of $P < .001$ was used as the level of statistical significance. Such an alpha level ensured that the overall error rate would not exceed 4.3%, which approximates the statistical error rate of 5% that is usually accepted for clinical studies.

Results

The mean values and SD for the hard tissue COGS analysis in American black adults are presented in Table 1. The mean values for white adults and the statistical level of significance between black and white adult means are shown in the last two columns. Table 2 shows the results of the soft tissue analysis. Table 3 compares the mean values and SD for hard tissues of black American males and females, and Table 4 compares the values for the soft tissues of black men and women.

The posterior cranial base (Ar-PTM) was significantly larger in the anteroposterior dimension in blacks than in whites. Among the horizontal and vertical skeletal and dental relations, the skeletal angle of facial convexity (N-A-Pg), maxillary skeletal protrusion (N-A), skeletal lower anterior face height (ANS-Me), upper posterior face height (PNS-N), upper anterior dental height (1/-NF), lower anterior dental height (1-MP), and lower pos-

terior dental height (6-MP) were significantly greater in black adults. Mandibular skeletal prognathism (N-B) and chin protrusion (N-Pg) were greater in black subjects than in white subjects, as indicated by less negative numbers, but these data were not statistically significant according to the criteria of this study. Similarly, the upper posterior dental height (6/-NF) was greater in black subjects, but not significantly.

Mandibular body length (Go-Pg) was significantly greater in blacks, whereas the ramus length (Ar-Go) was quite similar. Maxillary length (PNS-ANS) was somewhat longer in blacks, but not significantly. Chin depth (B-Pg) was significantly less in blacks than in whites. The only significantly different dental relationship between blacks and whites was that blacks had greater lower incisor proclination (1-MP). The similarity in dental relationships was most likely due to the selection criterion of ideal Angle class I occlusion.

Table 2 presents the results of the soft tissue analysis. The only statistically significant difference in overall facial form between black and white subjects was a greater soft tissue chin thickness, the distance between hard and soft tissue pogonion (Pg-Pg'). The greater soft tissue chin thickness may explain why the soft tissue facial contour angle (G-Sn-Pg') was not statistically different than that of white subjects, whereas the skeletal angle of fa-

Table 1. Skeletal and Dental Cephalometric Values: Blacks vs Whites

Measurement*	Description†	Black	SD‡	White	P§
Cranial base					
Posterior cranial base	Ar-PTM ($^{\circ}$ HP)	39.0	4.3	35.0	<.001
Anterior cranial base	PTM-N ($^{\circ}$ HP)	50.6	3.8	51.9	NS
Horizontal skeletal relations					
Facial convexity	N-A-Pg (angle)	10.2	5.8	3.3	<.001
Maxillary protrusion	N-A ($^{\circ}$ HP)	3.4	3.7	-1.0	<.001
Mandibular protrusion	N-B ($^{\circ}$ HP)	-3.0	6.9	-6.1	<.050
Chin protrusion	N-Pg ($^{\circ}$ HP)	-3.7	8.0	-5.4	NS
Vertical skeletal and dental relations					
Upper anterior face height	N-ANS (PHP)	54.1	3.2	52.4	<.010
Lower anterior face height	ANS-Me (PHP)	71.7	9.0	65.0	<.001
Upper posterior face height	PNS-N (PHP)	55.0	3.8	52.3	<.001
Mandibular plane angle	MP-HP (angle)	25.4	6.3	23.6	NS
Upper anterior dental height	1/-NF (PNF)	31.8	3.9	29.0	<.001
Lower anterior dental height	/1-MP (PMP)	47.6	5.0	42.9	<.001
Upper posterior dental height	6/-NF (PNF)	25.9	2.7	24.6	<.010
Lower posterior dental height	/6-MP (PMP)	37.4	3.6	34.0	<.001
Maxilla and mandible					
Maxillary length	PNS-ANS ($^{\circ}$ HP)	57.5	3.9	55.2	<.010
Mandibular ramus length	Ar-Go (linear)	50.5	4.8	49.4	NS
Mandibular body length	Go-Pg (linear)	85.6	5.9	79.0	<.001
Chin depth	B-Pg ($^{\circ}$ MP)	5.5	2.3	8.1	<.001
Gonial angle	Ar-Go-Me (angle)	119.7	7.1	120.6	NS
Dental relationships					
Wits analysis	A-B ($^{\circ}$ OP)	-1.0	4.2	-0.8	NS
Upper incisor inclination	1/-NF (angle)	113.4	7.5	111.8	NS
Lower incisor inclination	/1-MP (angle)	100.7	6.9	95.9	<.001
Upper occlusal plane	OP/-HP (angle)	7.1	4.9	6.7	NS
Lower occlusal plane	/OP-HP (angle)	-3.1	5.1	0.0	<.010

* Linear measurements are expressed in millimeters; angular measurements are expressed in degrees.

† $^{\circ}$ HP, $^{\circ}$ MP, $^{\circ}$ OP: measured parallel to the horizontal plane, mandibular plane, and occlusal plane, respectively. PHP, PNF, PMP: measured perpendicular to the horizontal plane, nasal floor, and mandibular plane, respectively.

‡ SD, standard deviation of the black sample.

§ P values <.05 are listed, but the level of significance set for this study was $P < .001$. NS, not significant.

cial convexity (N-A-Pg) was more acute in black subjects than in white subjects. There was a greater amount of midface protrusion (G-Sn) in blacks than in whites, but this difference was not significant according to the criteria of this study.

There were several differences in labial and nasal form between black and white subjects. The nasolabial angle (Cm-Sn-UL) was more acute and the upper lip length (Sn-Stm_s) was greater in black subjects. Upper and lower lip protrusion beyond the lower facial plane (Sn-Pg') were greater in black subjects, indicating greater soft tissue thickness in this region. The greater soft tissue thickness in the lower lip and in the chin contributed to the greater depth of the mentolabial sulcus (Si to Li-Pg') in blacks than in whites. The lower lip length (Stm_s-Me') was significantly greater in blacks. The mean figure for the interlabial gap was smaller in black subjects, because of a low incidence of lip incompetence in the sample studied. Nasal depth (RN-Sn), nasal projection (Sn-PN), and nasal pro-

file angle (Sn-N'-PN) were all significantly smaller in black than in white subjects.

Tables 3 and 4 demonstrate the sexual dimorphism found in black American adults. All values that were significantly different according to the criteria of this study showed that black men are larger than black women. Lower incisor inclination was greater in women than in men, but not significantly. Lower anterior and posterior dental height were significantly larger in men. In addition, mandibular ramus length and upper posterior face height were significantly larger in men. Lower lip length was significantly larger in men, but the vertical height ratio (G-Sn/Sn-Me') was not significantly different, indicating that vertical facial proportions were fairly similar among black men and women.

Discussion

The problems that can be identified when comparing cephalometric studies of black Americans

Table 2. Soft Tissue Cephalometric Values: Blacks vs Whites

Measurement*	Description†	Black	SD‡	White	P§
Facial form					
Facial convexity angle	G-Sn-Pg' (angle)	12.5	5.9	12.1	NS
Midface protrusion	G-Sn (°HP)	7.7	4.2	6.0	<.050
Lower face protrusion	G-Pg' (°HP)	1.1	8.3	0.0	NS
Vertical height ratio	G-Sn/Sn-Me' (PHP)	1.0	0.1	1.0	NS
Lower face-throat angle	Sn-Gn'-C (angle)	104.3	13.3	100.0	NS
Lower face height-depth ratio	Sn-Gn'/Gn'-C	1.4	0.3	1.2	<.050
Soft tissue chin thickness	Pg-Pg' (°HP)	15.2	2.5	12.2	<.001
Lip position and form					
Nasolabial angle	Cm-Sn-Ls (angle)	91.3	14.1	102.0	<.001
Upper incisor display	Stms-IS (PHP)	1.8	2.2	2.0	NS
Upper lip length	Sn-Stms (PHP)	25.9	3.0	20.0	<.001
Lower lip length	Stms-Me' (PHP)	51.5	6.0	45.9	<.001
Interlabial gap	Stms-Stmi (PHP)	0.4	1.1	2.0	<.001
Upper lip protrusion	Ls to (Sn-Pg')	8.6	1.8	2.8	<.001
Lower lip protrusion	Li to (Sn-Pg')	6.9	2.7	1.8	<.001
Mentolabial sulcus	Si to (Li-Pg')	-5.9	1.5	-4.0	<.001
Nasal analysis					
Nasal depth	Rn-Sn (°HP)	11.0	2.5	13.3	<.001
Nasal projection	Pn-Sn (°HP)	11.9	2.1	15.7	<.001
Nasal profile angle	Sn-N'-Pn (angle)	22.9	2.2	25.0	<.001

* Linear measurements are expressed in millimeters; angular measurements are expressed in degrees.

† °HP: Measured parallel to the horizontal plane; PHP: measured perpendicular to the horizontal plane.

‡ SD, standard deviation of the black sample.

§ P values <.05 are listed, but the level of significance set for this study was $P < .001$. NS, not significant.

are types of analysis used, age of the sample, sample size, selection criteria, statistical methods, definitions of clinical normalcy, definitions of the black racial designation, and variation by geographic distribution of blacks in the United States. Previous studies of cephalometric norms in American black adults have reported values for analyses other than the COGS analysis. Normal values for the COGS analysis are now available for white adults only. This cephalometric study of black American adults presents norms developed from the complete COGS analysis, which will be useful in diagnosis and treatment planning for orthognathic surgery in black American adults.

Most patients presenting for orthognathic surgery are young adults. Due to the processes of facial growth and development, cephalometric norms for children can be expected to differ from those of adults. Therefore, the cephalometric studies whose samples included predominantly or exclusively black children⁹⁻¹² may not be useful for prospective orthognathic surgery cases. Similarly, patients of advanced age may show changes due simply to the aging process such as loss of vertical dimension between the jaws caused by attrition and loss of teeth. In this study the sample was limited to young adults, with a mean age of 25.8 years.

Some reports of cephalometric norms in black subjects have methodological problems of sample

size. The study by Cotton et al¹² was based on only 20 subjects. Altemus^{10,11} selected the 80 adolescents with the most ideal dentitions from a group of 3,289. Kowalski et al,¹⁸ on the other hand, studied a large sample of 244 subjects. Our study used 33 suitable records found by reviewing approximately 15,000 inactive charts from the University of Pittsburgh School of Dental Medicine; this sample size was similar to most other studies that have determined cephalometric norms.

The criteria on which subjects for study were selected vary among the available reports. Most studies used accepted norms for occlusion, such as Angle class I, while others did not.¹⁸ Some studies selected for ideal alignment of the teeth and presence of all the permanent teeth,^{9,10} while others selected only Angle class I dentitions and do not specify whether open bite, deep bite, partial edentulism, or other abnormal conditions were excluded.^{19,20} However, most reports do exclude subjects who have suffered facial trauma or undergone facial surgery or orthodontic treatment. The present study used the information in a comprehensive dental school patient record to identify subjects that had an ideal dentition and occlusion.

Selection of subjects according to ideal dental occlusion in studies comparing black and white subjects tends to minimize differences in cephalometric dental measurements between the two groups.

Table 3. Skeletal and Dental Comparison: Black Men vs Black Women

Measurement*	Description†	Men	SD	Women	SD	P‡
Cranial base						
Posterior cranial base	Ar-PTM (°HP)	41.1	3.9	37.2	4.0	<.010
Anterior cranial base	PTM-N (°HP)	51.5	3.1	49.9	4.2	NS
Horizontal skeletal relations						
Facial convexity	N-A-Pg (angle)	9.9	6.3	10.4	5.5	NS
Maxillary protrusion	N-A (°HP)	3.2	3.4	3.5	4.0	NS
Mandibular protrusion	N-B (°HP)	-3.5	6.7	-2.6	7.2	NS
Chin protrusion	N-Pg (°HP)	-4.2	7.9	-3.3	8.4	NS
Vertical Skeletal and Dental Relations						
Upper anterior face height	N-ANS (PHP)	55.8	2.6	52.7	3.0	<.010
Lower anterior face height	ANS-Me (PHP)	74.5	10.9	69.3	6.5	NS
Upper posterior face height	PNS-N (PHP)	57.4	3.2	53.1	3.2	<.001
Mandibular plane angle	MP-HP (angle)	25.4	6.7	25.3	6.2	NS
Upper anterior dental height	I/-NF (PNF)	32.7	3.7	31.1	3.9	NS
Lower anterior dental height	/I-MP (PMP)	50.8	3.6	45.0	4.6	<.001
Upper posterior dental height	6/-NF (PNF)	27.1	1.8	25.0	2.9	<.050
Lower posterior dental height	/6-MP (PMP)	39.6	2.6	35.5	3.3	<.001
Maxilla and mandible						
Maxillary length	PNS-ANS (°HP)	58.8	3.9	56.4	3.8	NS
Mandibular ramus length	Ar-Go (linear)	53.5	4.4	48.1	3.7	<.001
Mandibular body length	Go-Pg (linear)	87.3	6.9	84.3	4.6	NS
Chin depth	B-PG (°MP)	5.9	2.8	5.1	1.9	NS
Gonial angle	Ar-Go-Me (angle)	120.4	8.9	119.1	5.4	NS
Dental relationships						
Wits analysis	A-B (°OP)	-2.5	5.1	0.3	2.8	NS
Upper incisor inclination	I/-NF (angle)	111.3	8.5	115.2	6.1	NS
Lower incisor inclination	/I-MP (angle)	97.7	6.0	103.2	6.6	<.050
Upper occlusal plane	OP/-HP (angle)	5.4	4.3	8.6	5.1	NS
Lower occlusal plane	/OP-HP (angle)	-4.6	4.2	-2.0	5.7	NS

* Linear measurements are expressed in millimeters; angular measurements are expressed in degrees.

† °HP, °MP, °OP: measured parallel to the horizontal plane, mandibular plane, and occlusal plane, respectively. PHP, PNF, PMP: measured perpendicular to the horizontal plane, nasal floor, or mandibular plane, respectively.

‡ *P* values <.05 are listed, but the level of significance set for this study was *P* < .001. NS, not significant.

Whether that selection also tends to minimize differences in skeletal measurements is unclear.

Many studies comparing black and white statistical norms use the mean values for white subjects developed by a different investigator, who may have used slightly different cephalometric landmark definitions and measurement techniques. Therefore, the two sets of results may not be directly comparable. Other studies use a control group of white subjects for comparison. Since cephalometric technique should be the same for both groups in the latter type of study, the results should be more comparable. In order to compare the norms of Burstone and colleagues^{1,2} for whites with those we found in blacks, concerted effort was made to duplicate their technique, including use of the same landmark definitions, the same computer program, and the same digitizing tablet used in their studies.

Baumrind and Frantz have shown that mean landmark identification error can be from 0.37 to 3.75 mm, depending on the particular landmark,²¹

and that measurement error can be 0.43 to 0.86 mm and 0.62° to 3.54° for linear and angular measurements, respectively, depending on the particular measurement.²² Landmark identification error can be reduced by the averaging of repeated measurements,²³ as in this study. In addition, landmark identification was repeatedly checked by two investigators, and differences in landmark identification were resolved with reference to the established landmark definitions. Measurement error is negligible when a computer makes measurements from digitized points; the pixel size of the Tektronix 4953 digitizing tablet used in this study was 0.39 mm². On the other hand, a new form of measurement error is introduced, caused by inaccuracy in touching the digitizing pen to the landmark identified on the acetate tracing. We minimized this error by averaging repeated measurements.

Studies which use multiple *t* tests to compare means for black and white samples may be subject to type I error. If the selected confidence interval

Table 4. Soft Tissue Comparison: Black Men vs Black Women

Measurement*	Description†	Men	SD	Women	SD	P‡
Facial form						
Facial convexity angle	G-SN-Pg' (angle)	13.3	6.7	12.0	5.4	NS
Midface protrusion	G-Sn (°HP)	8.4	3.6	7.2	4.7	NS
Lower face protrusion	G-Pg' (°HP)	1.1	8.1	1.1	8.7	NS
Vertical height ratio	G-Sn/Sn-Me' (PHP)	0.9	0.1	1.0	0.1	<.050
Lower face-throat angle	Sn-Gn'-C (angle)	111.2	12.1	99.4	12.2	<.050
Lower face height-depth ratio	Sn-Gn'/Gn'-C	1.5	0.3	1.3	0.4	NS
Soft tissue chin thickness	Pg-Pg' (°HP)	16.6	2.5	14.1	2.0	<.010
Lip position and form						
Nasolabial angle	Cm-Sn-Ls (angle)	96.3	14.8	87.7	12.8	NS
Upper incisor display	Stms-IS (PHP)	1.5	2.7	2.2	1.7	NS
Upper lip length	Sn-Stms (PHP)	27.1	2.1	25.0	3.4	NS
Lower lip length	Stms-Me' (PHP)	55.2	5.2	47.8	4.9	<.001
Interlabial gap	Stms-Stmi (PHP)	0.2	0.6	0.6	1.4	NS
Upper lip protrusion	LS to Sn-Pg'	8.7	2.3	8.7	1.3	NS
Lower lip protrusion	Li to Sn-Pg'	6.4	3.0	7.3	2.4	NS
Mentolabial sulcus	Si to Li-Pg'	-6.5	1.7	-5.5	1.3	NS
Nasal analysis						
Nasal depth	Rn-Sn (°HP)	12.3	2.6	10.1	2.0	<.010
Nasal projection	Pn-Sn (°HP)	11.8	1.2	12.0	2.6	NS
Nasal profile angle	Sn-N'-Pn (angle)	22.8	2.2	23.0	2.2	NS

* Linear measurements are expressed in millimeters; angular measurements are expressed in degrees.

† °HP: measured parallel to the horizontal plane. PHP: measured perpendicular to the horizontal plane.

‡ P values <.05 are listed, but the level of significance set for this study was $P < .001$. NS, not significant.

for the t tests is $P < .05$, then there is a 5% chance that the t test found a statistically significant difference between the two groups when there actually was none (type I error). If, for example, ten t tests are used, there is a 50% chance ($10 \times 5\%$) that one of the t tests made a type I error. The probability of type I error due to the 41 t tests used in this study would approach certainty. The Bonferroni method of dividing the overall significance level desired for the study ($P < .05$) by the number of t tests used can conservatively determine the proper significance level for each test (.001).²⁴ Therefore, choosing the $P < .001$ confidence interval in this study reduced to acceptable levels the probability (alpha level) that any one of the t tests used falsely reported statistical significance.

The mean values for each cephalometric measurement, whether statistically different from the value in whites or not, can be used for a cephalometric analysis specifically designed for black American adults. On the other hand, this rigorous statistical method allows for more valid comparison of the black American adult norms with those of white adults. Similarly, the values for each sex can be used, whether or not they are statistically different than those of the opposite sex.

The historical admixture of many peoples and races who now call themselves "black" in America makes biological definition of that group impossi-

ble. The dental charts designated as American blacks in this study were those that had any of the following words entered in the identifying data section of the chart: Black, Negro, Colored, Afro-American, or Bilalian (1 case). It appeared that most, if not all, of the patients filled out the identifying data section of the chart by themselves. Therefore, the black racial designation was self-selected by the subjects of this study. Most other cephalometric studies of American blacks did not address this question.

The question of ideal black facial aesthetics was not addressed in this study. Only Burstone's use of white adolescent subjects selected by artists, teachers, or housewives for their "acceptable" facial form in his studies of the integumental profile⁷ attempted to address facial aesthetics in a cephalometric study. The present study resembles most other studies of black American adults in that the dental occlusion was used as the main selection criterion. Thus the norms developed in this study describe the average facial structure of that portion of the black American adult population that has an ideal, presumably functional dental occlusion.

The sample used in this study was drawn from the adult black population of Pittsburgh. Whether the maxillofacial structure of American blacks from Pittsburgh is significantly different from that of black adults from other parts of the United States is

not known. It is evident from this study, though, that the general cephalometric features of this sample are comparable to those of other black populations studied in Washington, DC,⁹⁻¹¹ San Francisco,¹² Ann Arbor, MI,^{18,19} and St Louis.²⁰

Conclusions

Surgically useful rectilinear cephalometric norms for the diagnosis and treatment planning of orthognathic surgery in American black adults have been developed. The results of this study allow the following conclusions:

1. In black subjects, there is greater maxillary skeletal prognathism, greater skeletal lower anterior face height, greater upper posterior face height, greater proclination of the lower incisor, greater posterior cranial base length, greater mandibular body length, and greater skeletal facial convexity than in whites. In addition, the upper and lower anterior dental heights, upper and lower lip lengths, and soft tissue thickness of the lips and chin are greater in black subjects than in whites.

2. There is less nasal depth and projection, less bony chin depth, and a smaller nasolabial angle in black subjects than in white subjects.

3. In comparison with some other studies of black American adults, statistically significant maxillary dental proclination and mandibular prognathism were not found.

4. The facial structure of black men in general is larger than that of black women.

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